Calculus II

Tutorial Problems #6 24–28 Mar 2025

Topics:

- Evaluating iterated integrals
- Evaluating double integrals over rectangles
- Evaluating double integrals over non-rectangles
- Area by double integrals

Problem 1. Evaluate the iterated integral.

$$\int_1^4 \int_1^e \frac{\ln x}{xy}\, dx\, dy.$$

In Problems 2–4, evaluate the double integral over the given region R.

Problem 2.

$$\iint_R \left(\frac{\sqrt{x}}{y^2}\right) dA, \quad R: \quad 0 \le x \le 4, \quad 1 \le y \le 2.$$

Problem 3.

$$\iint_R \frac{xy^3}{x^2+1}\, dA, \quad R: \quad 0 \le x \le 1, \quad 0 \le y \le 2.$$

Problem 4.

$$\iint_R \frac{y}{x^2y^2+1}\, dA, \quad R: \quad 0 \le x \le 1, \quad 0 \le y \le 1.$$

Problem 5. *Maximizing a double integral.* What region R in the $xy-$ plane maximizes the value of

$$\iint_R (4 - x^2 - 2y^2)\, dA?$$

Justify your answer.

In Problems 6–7, sketch the region bounded by the given lines and curves. Then express the region's area as an iterated double integral and evaluate the integral.

Problem 6. The parabola $x = -y^2$ and the line $y = x + 2$.

Problem 7. The parabolas $x = y^2$ and $x = 2y - y^2$.

The integrals and sums of integrals in Problems 8–9 give the areas of regions in the $xy-$plane. Sketch each region, label each bounding curve with its equation, and give the coordinates of the points where the curves intersect. Then find the area of the region.

Problem 8.

$$\int_0^6 \int_{y^2/3}^{2y} dx\, dy.$$

Problem 9.

$$\int_0^{\pi/4} \int_{\sin x}^{\cos x} dy\, dx.$$

Problem 10. Find the average height of the paraboloid $z = x^2 + y^2$ over the square $0 \le x \le 2$, $0 \le y \le 2$.

END

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Answers to selected problems

1. $\ln 2$
2. $\frac{8}{3}$
3. $2 \ln 2$
4. $\frac{\pi}{4} - \frac{1}{2} \ln 2$
6. $\frac{9}{2}$.
7. $\frac{1}{3}$
8. 12
9. $\sqrt{2} - 1$
10. $\frac{8}{3}$.